

# Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

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# Table of Contents

- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients
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# Motivation

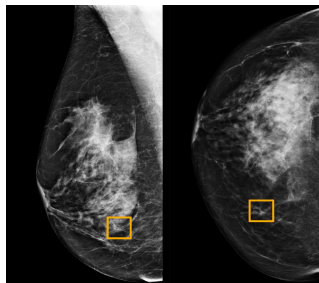
High stakes

Consider these situations:

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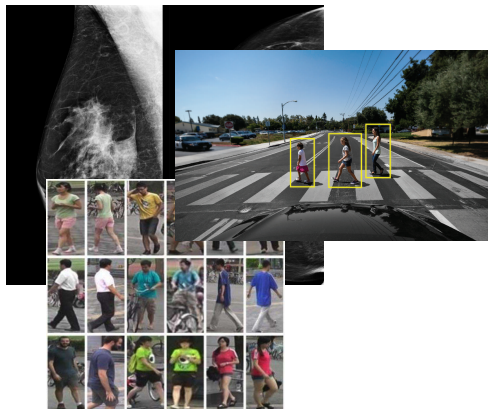
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# Motivation

High stakes

Consider these situations:



## Straight to the point



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

We must **understand** the  
behaviour of these models.

## Step by step

# Computation, Computer Vision and AI



## Explainable AI



## Thesis objectives

# Computation



Better known as *Computer Science*.

## Study of:

- Algorithms.
- Data structures.
- Design of hardware and software.

# Computation, Computer Vision and AI

## Computer Vision

Replication of human vision capabilities.

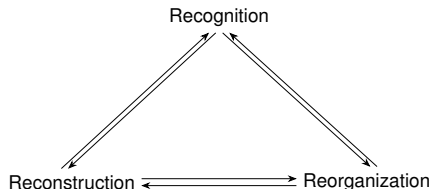
# Computation, Computer Vision and AI

## Computer Vision

Replication of human vision capabilities.

Three fundamental tasks[1]:

- Recognition.
- Reconstruction.
- Reorganization.



# Computation, Computer Vision and AI

## Artificial Intelligence

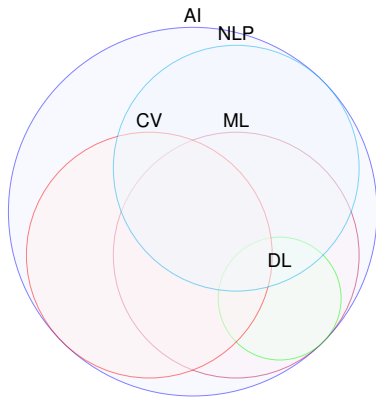
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# Artificial Intelligence

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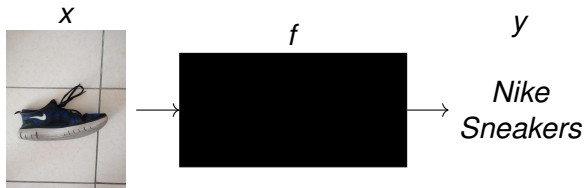
Subfields:

- Machine Learning (*ML*) & Deep Learning (*DL*).
- Computer Vision(*CV*).
- Natural Language Processing (*NLP*).
- Robotics.



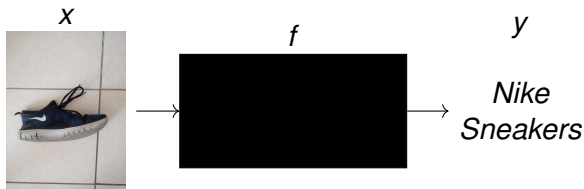
# Explainable AI

We are interested in understanding models, behaving like a black box model:



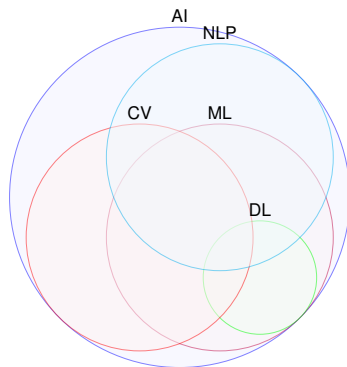
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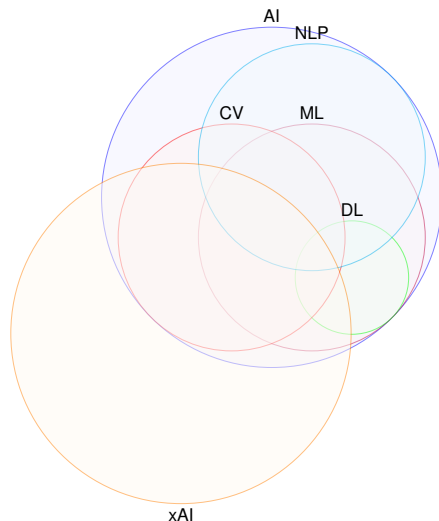


We want to *know why*  $f(x) \rightarrow y$

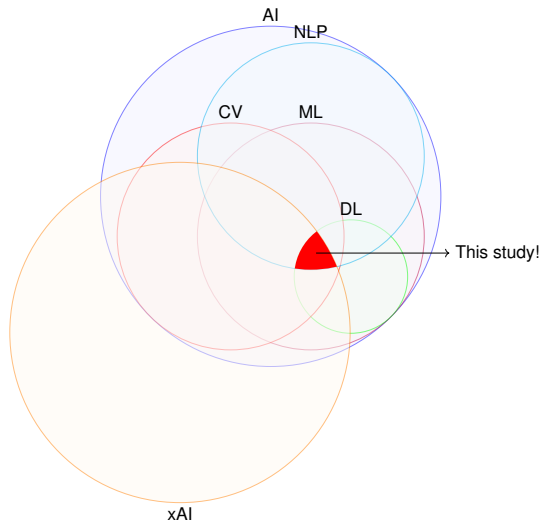
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# Thesis Objectives

**Improvement of recognition and interpretable properties of model predictions.**

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## **Improvement of recognition and interpretable properties of model predictions.**

In particular:

- Development of low cost/complexity explainability approaches.
- Establishment of a fixed evaluation protocol.
- Differentiation of human based and machine explanations.

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# Preliminaries

## Desired data of Interpretability Study [4]

We ask questions regarding *black box models*.

- How does it work?
- How safe is it?
- Who is accountable for it?
- Who benefits from it?
- Who uses it?

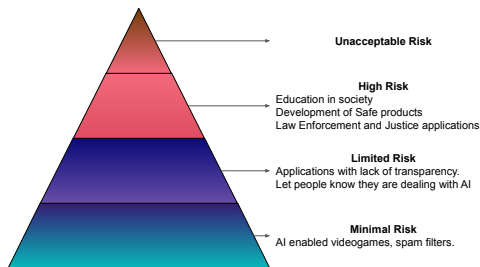
**Accepted AI systems within society must answer this.**

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Regulation planned with the European AI act[3]

**Accepted AI systems within society must answer this.**

# Image Recognition Models

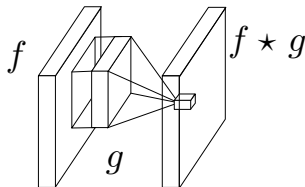
## Convolutional Neural Networks (CNN)

Based on the **convolution operation**.

A representation  $f \star g$  is computed for a feature map  $f$  and a kernel  $g$ .

First approach with *Neocognitron*

[5]

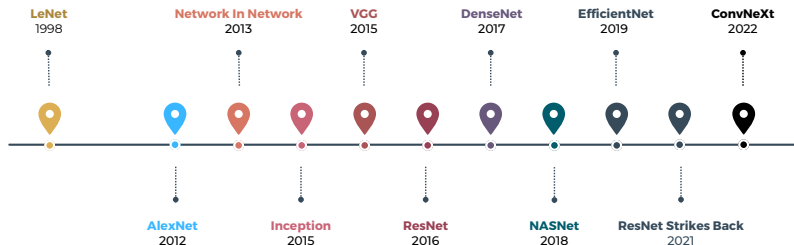
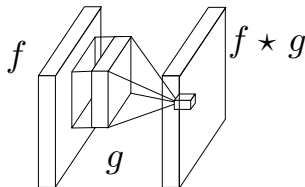


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# Image Recognition Models

## Convolutional Neural Networks(CNN)

Training deep models is difficult, gradient information *is* lost with depth.

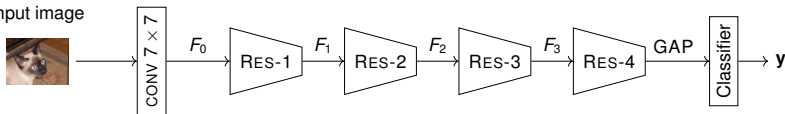
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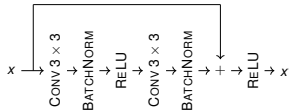
Training deep models is difficult, gradient information *is* lost with depth.

A model surviving the test of time: *ResNet* [6].

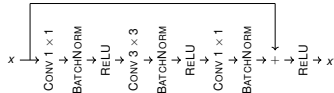
Input image



### Basic Block



### Bottleneck Block



# Image Recognition Models

## Self Attention Architectures

### Transformers

*Attention:* compute the relevance of each element relative to others in an embedding.

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

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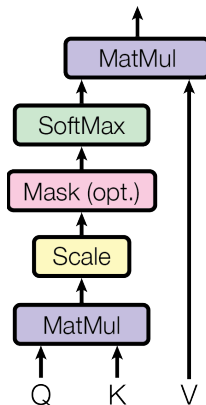
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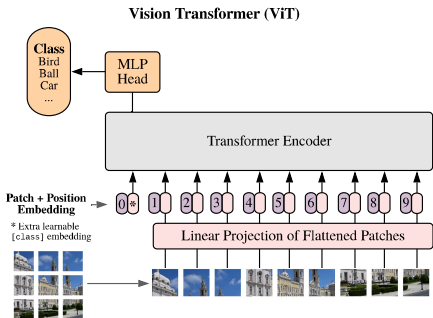
An abstract representation of a class is computed, updated with information across the embedding [7].



# Image Recognition Models

## Self Attention Architectures

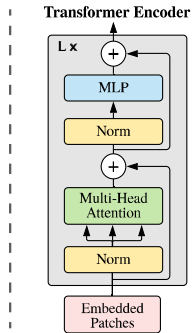
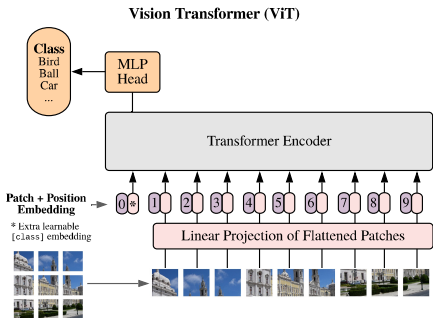
A modification of transformers to vision, the *Vision Transformer* (ViT)  
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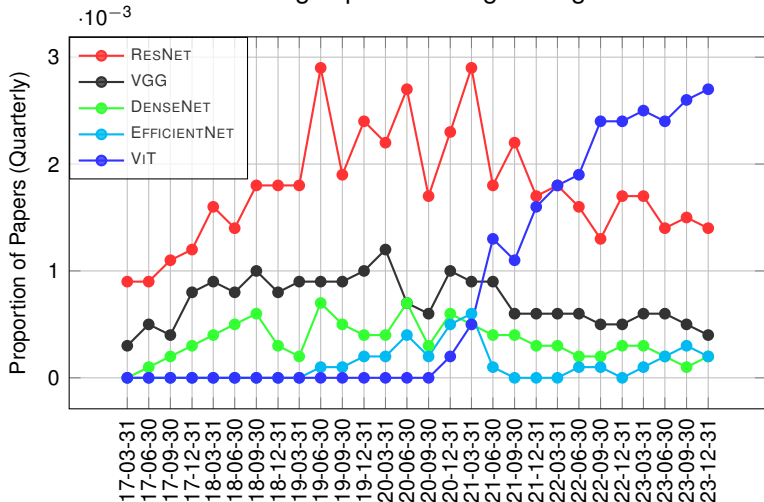
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# Image Recognition Models

## The Current Landscape

Transformers had a strong impact on image recognition:



# Interpretability

## Two dimension grouping

# Interpretability

## Two dimension grouping

### Transparency

Approaches requiring changes to an architecture or training process.

### Post-Hoc Interpretability

Approaches utilizing information contained within a model.

# Interpretability

## Class Activation Methods

Semantic information is found in deeper stages of a model [9].



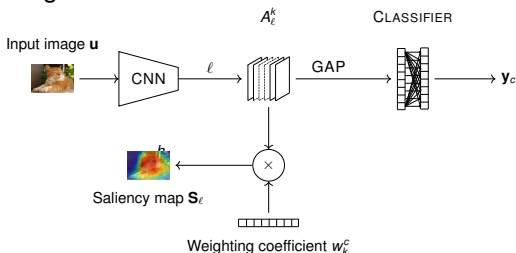
# Interpretability

## Class Activation Methods

Semantic information is found in deeper stages of a model [9].



Salient regions are highlighted for an input image:



Governed by equation:

$$S_\ell^c(\mathbf{u}) := h \left( \sum_k w_k^c A_\ell^k \right) \quad (2)$$

# Interpretability

## Evaluating Interpretability

Saliency: what is relevant towards a prediction.

### Image Classification Approach

An explanation should demonstrate similar predictive properties to its query:

Input image. ( $I$ )



$$P_i^c = 0.8756$$

Explanation Map. ( $E^c$ )



$$O_i^c = 0.7442$$

# Interpretability

## Evaluating Interpretability

Saliency: what is relevant towards a prediction.

### Image Classification Approach

An explanation should demonstrate similar predictive properties to its query:

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$$P_i^c = 0.8756$$

- Prediction loss when using an explanation. (Average Drop (AD) [10])
- Ratio of explanations being better than the input image. (Average Increase (AI) [10])
- Reaction to saliency guided modifications (Insertion/Deletion (Ins/Del) [11])

Explanation Map. ( $E^c$ )



$$O_i^c = 0.7442$$

# Interpretability

## Evaluating Interpretability

Saliency: what is relevant towards a prediction.

### **Weakly Supervised Object Localization Approach**

A saliency map should be good to localize the object of interest for an example:

Input image. ( $I$ )



Saliency Map. ( $E^c$ )



# Interpretability

## Evaluating Interpretability

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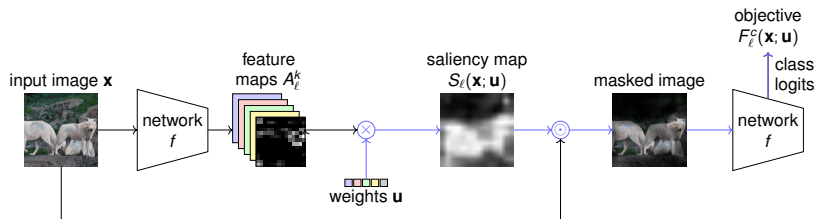
- Relationships between the bounding boxes (OM, LE,  $F_1$ ).
- Overlap between object mask and bounding box (BA).
- Localization of most salient points and fraction of them in the bounding box (EP, SP).
- Penalty of the size of the bounding box and image size, regarding prediction (SM).



# Preliminaries

## Premise

*Is it possible to optimize the weighting coefficient  $w_k^c$  to obtain a saliency map with the highest probability of prediction?*



# Preliminaries

## Contributions

### Opti-CAM

Maximization of predicted probability of an explanation.

### Average Drop

Evaluation metric designed to enhance comparisons.

# Opti-CAM

## Motivation

- **Combination of Feature Maps**

To compute  $w_k^c$ , Score-CAM [12] considers feature maps independently.

*What about a linear combination?*

$$F(\mathbf{w}) := f \left( \mathbf{u} \odot n \left( \text{up} \left( \sum_k w_k A_l^k \right) \right) \right)_c$$

F is optimizable with respect to  $\mathbf{w}$  and an initial  $w_0$  via backpropagation.

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### ● Masking-based Saliency Saliency maps produced by optimization in the input space [13, 14]

The optimization usually takes form of:

$$S^c(\mathbf{u}) := \underset{\mathbf{m} \in M}{\operatorname{argmax}} f(\mathbf{u} \odot n(\text{up}(\mathbf{m})))_c + \delta R(\mathbf{m})$$

# Opti-CAM

## Optimization

**Formulation** Assuming channel weights  $w_k := \text{softmax}(\mathbf{u})_k$ , the saliency map  $S_\ell$  is a function of both an input image  $x$  and a variable  $\mathbf{u}$ :

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

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Find vector  $\mathbf{u}^*$  maximizing confidence for class  $c$ , with an input  $\mathbf{x}$  masked by Saliency Map  $S_\ell(\mathbf{x}; \mathbf{u}^*)$ :

$$\mathbf{u}^* := \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}^*), \quad (4)$$

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The objective function:

$$F_\ell^c(\mathbf{x}; \mathbf{u}) := g_c(f(\mathbf{x} \odot S_\ell(\mathbf{x}; \mathbf{u}))) \quad (5)$$

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The Saliency Map then becomes:

$$S_\ell^c(\mathbf{x}) := S_\ell(\mathbf{x}; \mathbf{u}^*) = S_\ell(\mathbf{x}; \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u})) \quad (6)$$

# Average Gain

## Motivation

Existing metrics for measuring interpretability ( $AD\downarrow$ ,  $AI\uparrow$ ), can be fooled.

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### Fake-CAM [15]

What happens if you provide a Saliency Map where all the pixels *but-one* are 1?

Input



$$p_{389} = 0.874$$

Explanation



$$o_{389} = 0.822$$

- Drop = 5.95%
- Increase = 0%

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Therefore, we don't drop much predictive power, *but*, we don't gain confidence in our prediction.

# Average Gain

## Definition

With the notation on slide (22):

$$AG(\%) := \frac{1}{N} \sum_{i=1}^N \frac{[o_i^c - p_i^c]_+}{1 - p_i^c} \cdot 100 \quad (7)$$

Measuring the increase in predictive power when masking an input.

# Experimental Set-Up

## Implementation

- All images are set to  $224 \times 224 \times 3$ .
- Adam [16] optimizer, for 100 iterations.
- Cross Entropy.
- Stop optimizing if loss is less than  $10^{-10}$ .

---

<sup>1</sup>Pytorch Model Zoo <https://pytorch.org/vision/0.8/models.html>

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## Datasets and Models

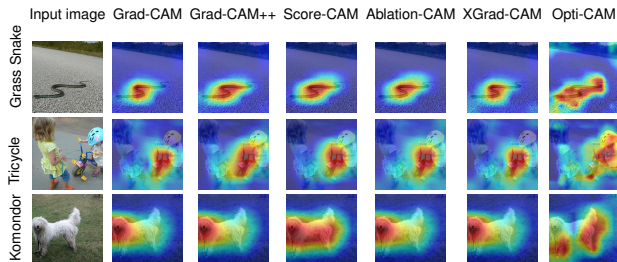
- **ILSVRC** Validation splits of ImageNet ([17, 18]), 50k images, over 1k classes.
- **Medical Data** Chest X-ray images with 3 diseases [19], endoscopy images for pathologies, findings and landmarks [20].
- **Networks** ResNet-50 [6], and VGG [21]. <sup>1</sup>ViT-B (16-224) [8] and DeiT-B (16-224) [22] <sup>2</sup>.

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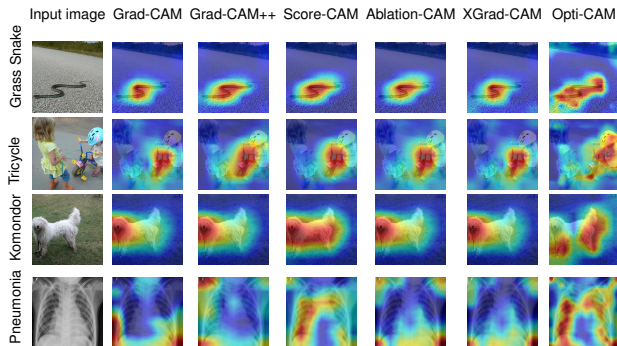
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## Qualitative Evaluation



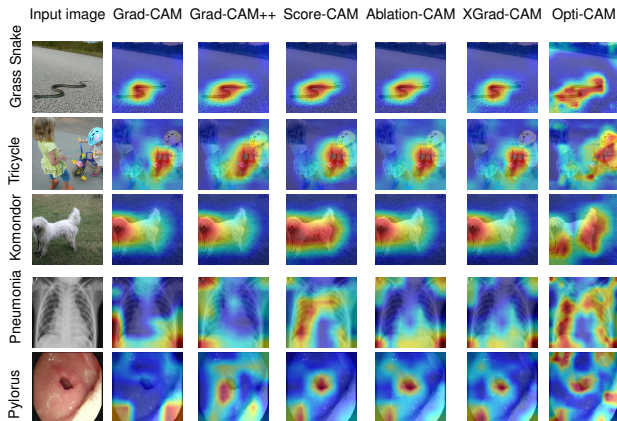
# Results

## Qualitative Evaluation



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## Qualitative Evaluation



# Results

## Quantitative Evaluation

### Classification Experiments:

| METHOD         | RESNET50   |             |             |             |             |        | VGG16      |             |             |             |            |       |
|----------------|------------|-------------|-------------|-------------|-------------|--------|------------|-------------|-------------|-------------|------------|-------|
|                | AD ↓       | AG ↑        | AI ↑        | I ↑         | D ↓         | T(s)   | AD ↓       | AG ↑        | AI ↑        | I ↑         | D ↓        | T(s)  |
| Fake-CAM       | 0.8        | 1.6         | 46.0        | 50.7        | 28.1        | 0.00   | 0.5        | 0.6         | 42.6        | 46.1        | 26.9       | 0.00  |
| Grad-CAM       | 12.2       | 17.6        | 44.4        | 66.3        | 14.7        | 0.03   | 14.2       | 14.7        | 40.6        | <b>64.1</b> | 11.6       | 0.02  |
| Grad-CAM++     | 12.9       | 16.0        | 42.1        | 66.0        | 14.7        | 0.03   | 17.1       | 10.2        | 33.4        | 62.9        | 12.2       | 0.02  |
| Score-CAM      | 8.6        | 26.6        | 56.7        | 65.7        | 16.3        | 15.22  | 13.5       | 15.6        | 41.7        | 62.5        | 12.1       | 3.11  |
| Ablation-CAM   | 12.5       | 16.4        | 42.8        | 65.9        | 14.6        | 18.26  | 15.5       | 12.6        | 36.9        | 63.8        | 11.4       | 2.98  |
| XGrad-CAM      | 12.2       | 17.6        | 44.4        | 66.3        | 14.7        | 0.03   | 13.8       | 14.8        | 41.2        | <b>64.1</b> | 11.7       | 0.02  |
| Layer-CAM      | 15.6       | 15.0        | 38.8        | 67.0        | <b>14.2</b> | 0.08   | 48.9       | 3.1         | 13.5        | 58.3        | <b>6.4</b> | 0.07  |
| ExPerturbation | 38.1       | 9.5         | 22.5        | <b>70.7</b> | 15.0        | 152.96 | 43.0       | 7.1         | 20.5        | 61.1        | 15.0       | 83.20 |
| Opti-CAM       | <b>1.5</b> | <b>68.8</b> | <b>92.8</b> | 62.0        | 19.7        | 4.15   | <b>1.3</b> | <b>71.2</b> | <b>92.7</b> | 59.2        | 11.0       | 3.94  |

# Results

## Quantitative Evaluation

### Localization Experiments:

| METHOD         | RESNET50    |             |             |             |             |             |             |  | VGG16       |             |             |             |             |             |             |  |
|----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|
|                | OM↓         | LE↓         | F1↑         | BA↑         | SP↑         | EP↑         | SM↓         |  | OM↓         | LE↓         | F1↑         | BA↑         | SP↑         | EP↑         | SM↓         |  |
| Fake-CAM       | 63.6        | 54.0        | 57.7        | 47.9        | 99.8        | 28.5        | 0.98        |  | 64.7        | 54.0        | 57.7        | 47.9        | 99.8        | 28.5        | 1.07        |  |
| Grad-CAM       | 72.9        | 65.8        | 49.8        | <b>56.2</b> | 69.8        | 33.3        | 1.30        |  | 71.1        | 62.3        | 42.0        | 54.2        | 64.8        | 32.0        | 1.39        |  |
| Grad-CAM++     | 73.1        | 66.1        | <b>50.4</b> | <b>56.2</b> | 69.9        | 33.1        | 1.29        |  | 70.8        | 61.9        | 44.3        | 55.2        | 66.2        | 32.3        | 1.38        |  |
| Score-CAM      | <b>72.2</b> | 64.9        | 49.6        | 54.5        | 68.7        | 32.4        | <b>1.25</b> |  | 71.2        | 62.5        | <b>45.3</b> | <b>58.5</b> | <b>68.2</b> | 33.4        | 1.40        |  |
| Ablation-CAM   | 72.8        | 65.7        | 50.2        | 56.1        | 69.9        | 33.1        | 1.26        |  | 71.3        | 62.6        | 43.2        | 56.2        | 65.7        | 32.7        | 1.39        |  |
| XGrad-CAM      | 72.9        | 65.8        | 49.8        | <b>56.2</b> | 69.8        | 33.3        | 1.30        |  | 70.8        | 62.0        | 41.9        | 53.5        | 64.4        | 31.6        | 1.41        |  |
| Layer-CAM      | 73.1        | 66.0        | 50.1        | 55.5        | <b>70.0</b> | 33.0        | 1.29        |  | 70.5        | 61.5        | 28.0        | 54.7        | 65.0        | 32.4        | 1.45        |  |
| ExPerturbation | 73.6        | 66.6        | 37.5        | 44.2        | 64.8        | <b>38.2</b> | 1.59        |  | 74.1        | 66.4        | 37.8        | 43.3        | 62.7        | <b>36.1</b> | 1.74        |  |
| Opti-CAM       | <b>72.2</b> | <b>64.8</b> | 47.3        | 49.2        | 59.4        | 30.5        | 1.34        |  | <b>69.1</b> | <b>59.9</b> | 44.1        | 51.2        | 61.4        | 30.7        | <b>1.34</b> |  |

# Conclusions

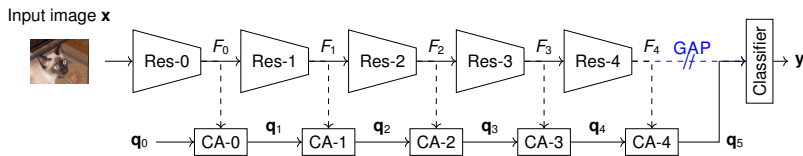
# Table of Contents

- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
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# Preliminaries

## Premise

*The [CLS] token in transformers updates an abstract representation via attention with patch embeddings across the depth of a model. Is it possible to train a similar class representation on CNNs?*



# Preliminaries

## Contributions

- **CAM is Attention**

We demonstrate that CAM behaves in a similar manner as attention.

- **Cross Attention: [CLS] on CNNs**

We train an abstract class representation using a stream, alongside standard CNNs.

- **CAM on more architectures**

CAM is usually validated only on ResNet and VGG.  
Extension to ConvNeXt.

# CAM is Attention

## Derivation

Let's freshen up:

Asuming  $\mathbf{F}_\ell \in \mathbb{R}^{p_\ell \times d_\ell}$  a reshaping of  $\mathbf{F}_\ell$ , and  $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$  the CLS token<sup>3</sup>.

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Attention is defined as:

$$\mathbf{a} = \mathbf{A}^\top = \text{softmax} \left( \frac{\mathbf{F}_\ell \mathbf{q}_\ell}{\sqrt{d_\ell}} \right). \quad (8)$$

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Upon replacing  $\mathbf{q}_\ell$  with an arbitrary vector  $\alpha \in \mathbb{R}^{d_\ell}$

$$\mathbf{a} = h_\ell(\mathbf{F}_\ell \alpha) = h_\ell \left( \sum_k \alpha_k \mathbf{f}_\ell^k \right). \quad (9)$$

With the activation function  $h_\ell(\mathbf{x}) = \text{softmax}(\mathbf{x}/\sqrt{d_\ell})$

---

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# CAM is Attention

## Pooling or Masking

We intend to integrate attention into a model, such that prediction and explanation are linked.

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**Cross Attention:**

$$\text{CA}_\ell(\mathbf{q}_\ell, F_\ell) := F^\top \mathbf{a} = F_\ell^\top h_\ell(F_\ell \mathbf{q}_\ell) \in \mathbb{R}_\ell^d \quad (10)$$

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Which, reshaping the feature matrix as  $F_\ell^\top = (\phi_\ell^1 \dots \phi_\ell^{p_\ell})$ , a pooled representation can be expressed, via attention vector

$\mathbf{a} = (a_1, \dots, a_{p_\ell})$ :

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A pooling similar to GAP, masked in a fashion similar to CAM.

# Cross Attention Stream

Given a model  $f$  we consider locations where critical operations take place:

$$f = g \circ \text{GAP} \circ f_L \circ \dots f_0 \quad (12)$$

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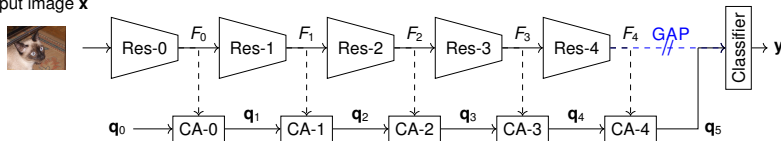
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Visualized as:

Input image  $\mathbf{x}$



# Experimental Set-Up

## Training and models

- Training and evaluation in the train and val sets of ILSVRC 2012 ([17, 18]) respectively.
- Training using ResNet 50 original procedure (90 epochs, SGD optimization [23], lr decay every 90 epochs).<sup>4</sup>
- Trained models ResNet (18-50), ConvNeXt (Small, Base).
- Pascal-VOC 2012[24] and CUB[25] extensions. Finetuning from ImageNet models.

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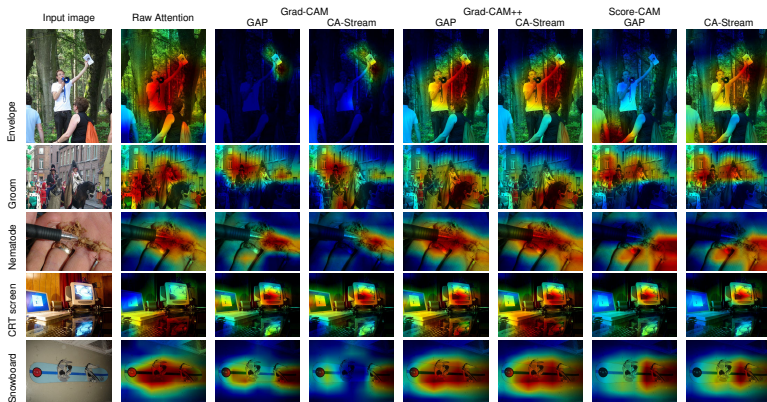
## Evaluation

- Evaluation using the whole image (not center crop), testing a bit of OOD.
- No localization evaluation, lessons learnt from Opti-CAM.
- Raw Attention used for OOD data, highlights on unseen data.
- Ablation on parameters and design choices.

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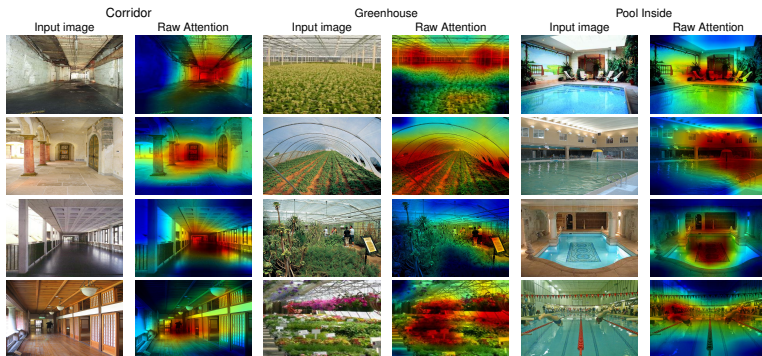
# Results

## Qualitative Experiments



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# Results

## Quantitative Experiments

| NETWORK    | METHOD     | POOL | AD↓          | AG↑          | AI↑          | I↑           | D↓           |
|------------|------------|------|--------------|--------------|--------------|--------------|--------------|
| RESNET-18  | Grad-CAM   | GAP  | 17.64        | 12.73        | 41.21        | 63.13        | <b>10.66</b> |
|            |            | CA   | <b>16.99</b> | <b>17.22</b> | <b>44.95</b> | <b>65.94</b> | 10.68        |
|            | Grad-CAM++ | GAP  | 19.05        | 11.16        | 37.99        | 62.80        | <b>10.75</b> |
|            |            | CA   | <b>19.02</b> | <b>14.76</b> | <b>40.82</b> | <b>65.53</b> | 10.82        |
|            | Score-CAM  | GAP  | 13.64        | 12.98        | 44.53        | 62.56        | <b>11.37</b> |
|            |            | CA   | <b>11.53</b> | <b>18.12</b> | <b>50.32</b> | <b>65.33</b> | 11.51        |
| CONVNEXT-S | Grad-CAM   | GAP  | 42.99        | 1.69         | 12.60        | 48.42        | <b>30.12</b> |
|            |            | CA   | <b>22.09</b> | <b>14.91</b> | <b>32.65</b> | <b>84.82</b> | 43.02        |
|            | Grad-CAM++ | GAP  | 56.42        | 1.32         | 10.35        | 48.28        | <b>33.41</b> |
|            |            | CA   | <b>51.87</b> | <b>9.40</b>  | <b>20.55</b> | <b>84.28</b> | 52.58        |
|            | Score-CAM  | GAP  | 74.79        | 1.29         | 10.10        | 47.40        | <b>38.21</b> |
|            |            | CA   | <b>64.21</b> | <b>8.81</b>  | <b>18.96</b> | <b>82.92</b> | 57.46        |

# Results

## Quantitative Experiments

| NETWORK    | METHOD     | POOL | AD↓          | AG↑          | AI↑          | I↑           | D↓           |
|------------|------------|------|--------------|--------------|--------------|--------------|--------------|
| RESNET-50  | Grad-CAM   | GAP  | 13.04        | 17.56        | 44.47        | 72.57        | <b>13.24</b> |
|            |            | CA   | <b>12.54</b> | <b>22.67</b> | <b>48.56</b> | <b>75.53</b> | 13.50        |
|            | Grad-CAM++ | GAP  | <b>13.79</b> | 15.87        | 42.08        | 72.32        | <b>13.33</b> |
|            |            | CA   | 13.99        | <b>19.29</b> | <b>44.60</b> | <b>75.21</b> | 13.78        |
|            | Score-CAM  | GAP  | 8.83         | 17.97        | 48.46        | 71.99        | <b>14.31</b> |
|            |            | CA   | <b>7.09</b>  | <b>23.65</b> | <b>54.20</b> | <b>74.91</b> | 14.68        |
| CONVNEXT-B | Grad-CAM   | GAP  | 33.72        | 2.43         | 15.25        | 52.85        | <b>29.57</b> |
|            |            | CA   | <b>19.45</b> | <b>13.96</b> | <b>32.89</b> | <b>86.38</b> | 45.29        |
|            | Grad-CAM++ | GAP  | <b>34.01</b> | 2.37         | 15.60        | 52.83        | <b>29.17</b> |
|            |            | CA   | 36.69        | <b>8.00</b>  | <b>21.95</b> | <b>85.39</b> | 53.42        |
|            | Score-CAM  | GAP  | 43.55        | 2.23         | 15.67        | 50.96        | <b>39.49</b> |
|            |            | CA   | <b>23.51</b> | <b>11.04</b> | <b>27.35</b> | <b>83.41</b> | 60.53        |

# Conclusions



# Table of Contents

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- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
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